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Upper Carboniferous Palæomagnetic pole from the stable Saharan Craton and Gondwana reconstructions

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ABSTRACT—A palæomagnetic study was conducted at different stratigraphical positions in rocks of Upper Carboniferous age (Bashkirian) in the Illizi Basin (Algeria). Three distinct remanence directions were isolated in these rocks. The first direction is associated with a Cenozoic magnetic overprint. The second direction yields a palæomagnetic pole (40.3°S, 64.3°E) that is similar to previously published African Permian poles and is interpreted as a Permian remagnetisation. The third direction gives a new Late Carboniferous (Bashkirian) African palæomagnetic pole (28.2°S, 55.5°E, $K = 207$, $A_{95} = 3.4°$). This pole is in a good agreement with previously published Saharan poles for the neighbouring periods. The combination of the new given datum with the previous Carboniferous poles of Gondwana shows that the reconstruction of Gondwana should be based on the parameters of Ricou *et al.* (1990) rather than on those of the classical du Toit (1937) model. © 2001 Elsevier Science Limited. All rights reserved.

RÉSUMÉ—L'étude paléomagnétique de différents niveaux d'âge carbonifère supérieur (Bashkirien) du bassin d'Illizi (Algérie) a permis de mettre en évidence une Aimantation Rémanente Caractéristique ayant selon les sites, trois orientations différentes. La première est associée à une réaimantation cénozoïque. La seconde, qui donne un pôle paléomagnétique (40.3°S, 64.3°E) très proche des pôles permiens africains précédemment déterminés, correspond probablement à une réaimantation permienne. La troisième direction permet de définir un nouveau pôle paléomagnétique africain d'âge bashkirien (28.2°S, 55.5°E, $K = 207$, $A_{95} = 3.4°$). Ce résultat est en bon accord avec les autres pôles sahariens pour des périodes voisines. La comparaison de cette nouvelle donnée avec les pôles carbonifères du Gondwana montre que la reconstruction du Gondwana doit être basée sur les paramètres de Ricou *et al.* (1990) au lieu de ceux du modèle classique de du Toit (1937). © 2001 Elsevier Science Limited. All rights reserved.

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INTRODUCTION

Despite numerous palæomagnetic studies devoted to Gondwana, poorly determined segments in its apparent polar wander path (APWP) still exist. Indeed, the widespread remagnetisation, the large uncertainty in the age of some studied rocks and the mediocre quality of some of the palæomagnetic data lead to conflicting results (Bachtadse and Briden, 1990; Henry *et al.*, 1992) and to the existence of various APWPs for Gondwana (van der Voo, 1993), particularly during Mid and Upper Palæozoic times (Smith *et al.*, 1994). Therefore, the model of convergence between Gondwana and Laurussia, and the continental readjustment which occurred after the Hercynian collision, are still debated (van der Voo and French, 1974; Irving, 1977;

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van der Voo, 1993; Chen *et al.*, 1994; Besse *et al.*, 1996; Torcq *et al.*, 1997).

To solve this problem, it is necessary to improve the precision of the Gondwana APWP by determining additional palæomagnetic poles. Additional palæomagnetic investigations of Palæozoic rocks have to be undertaken wherever possible within well-dated geological sections in a stable area such as the Sahara Craton (stable Africa). The well-studied Upper Carboniferous (Bashkirian) beds cropping in the Illizi Basin seemed to be particularly favourable for such a study.

GEOLOGICAL SETTING

The studied levels consist of the upper part of the 'Oubarakat Formation' and the lower part of the 'El-Adeb Larache Formation'. Two layers delimit them: the limestone of Oubarakat at the southern border and a gypsum level at the northern border (Fig. 1). The thickness of the Upper Oubarakat Formation is about 35 m. It consists of grey calcareous sandstone which contains some micro-organisms (zone 8: Massa and Vachard, 1979), such as *Eostaffela sp.*, *Eostaffelina sp.*, and *Millerella sp.*, attributed to the Lower Bashkirian (Legrand-Blain, 1983). The base of the 'El-Adeb Larache Formation' consists of blue limestone and white marls. The microfauna with *Pseudostaffela antiqua* (zone 9: Massa and Vachard, 1979) characterise the Lower Bashkirian age (~315 Ma: Odin, 1994).

The beds just above the bottom of the 'El-Adeb Larache Formation' are frequently red-coloured marls lacking fauna. Since the position of these beds is between the blue limestone of Lower Bashkirian and the well-dated Moscovian Beds (Legrand-Blain, 1980) of the Upper 'El-Adeb Larache Formation', their age is probably Upper Bashkirian (Legrand-Blain, 1983).

The Bashkirian Beds are situated within a structurally stable area and they appear as a tabular formation with dips that range from 0° to 4° (often $<4°$). This weak tilting is probably of Permian and/or Mesozoic age. Fabre (1988) points out that the most important tectonic phases affecting the Palæozoic formations of the eastern part of the Algerian Sahara occurred during the Permian. However, Mesozoic formations of the Illizi Basin also have a moderate regional dip (Kies *et al.*, 1995). The palæomagnetic data are not significantly different before and after dip correction. Therefore, all the palæomagnetic results will be presented in this paper only after dip correction.

SAMPLING AND ANALYSIS PROCEDURE

Two hundred and seventy oriented samples were collected at 24 sites scattered over a 10 km radius of the central point, about (27.5°N, 8.8°E; Fig. 1). The sampling sites are distributed throughout different stratigraphic levels of the studied formation (Fig. 1), sites (12–24) in the Lower Bashkirian Formation and sites (1–11) in the Upper Bashkirian Formation. One to three specimens were cut from each core in order to have additional specimens for pilot studies and rock magnetic analysis. The specimens were put in a zero field for at least one month, prior to any demagnetisation analysis, in order to reduce existing viscous components. The natural remanent magnetisation (NRM) of the specimens was measured using a JR4 spinner magnetometer (GEOFYZIKA, Brno). Several pilot specimens were subjected to a step-wise alternating field (up to 180×10^{-3} T) and thermal demagnetisation (up to 660°C) to characterise their magnetic behaviour.

All the samples (pale facies) belonging to a few sites of the lower part of the formation exhibit very low NRM intensities. The NRM of the pilot specimens collected from these sites became immeasurable after the first demagnetisation step (thermal or alternating field). All of the 80 samples from these sites have, therefore, been discarded.

For the other sites, the pilot specimen demagnetisation shows that the alternating field procedure did not result in complete demagnetisation. Therefore, the thermal demagnetisation was chosen for the remaining specimens. The NRM of the 190 samples was measured, but only 150 were thermally demagnetised; the remaining samples have very low NRM intensities and were not studied further. To obtain precise information on the magnetisation evolution, the thermal demagnetisation steps were carefully selected in order to fully resolve vector components (increments range from 150°C at the lowest temperature to 10°C at the highest temperatures).

During thermal demagnetisation, the magnetisation directions of the samples, after elimination of the weak viscous component (component A), show two types of behaviour. The first is a stable magnetic direction corresponding to the characteristic remanent magnetisation (ChRM). The second is a trajectory on the projection sphere along a great circle path that shows a superposition of unblocking temperature spectra from at least two components.

Demagnetisation data analysis was carried out using classical methods. The directions of the magnetisation components were presented on orthogonal vector plots (Wilson and Everitt, 1963; Zijderveld, 1967). The remaining vectors and vectorial differences of the magnetisation were plotted on equal area projections. The remagnetisation circles method (Halls, 1976; 1978; McFadden and McElhinny, 1988) was also used. The mean directions of the different components have been computed using Fisher (1953) statistics.

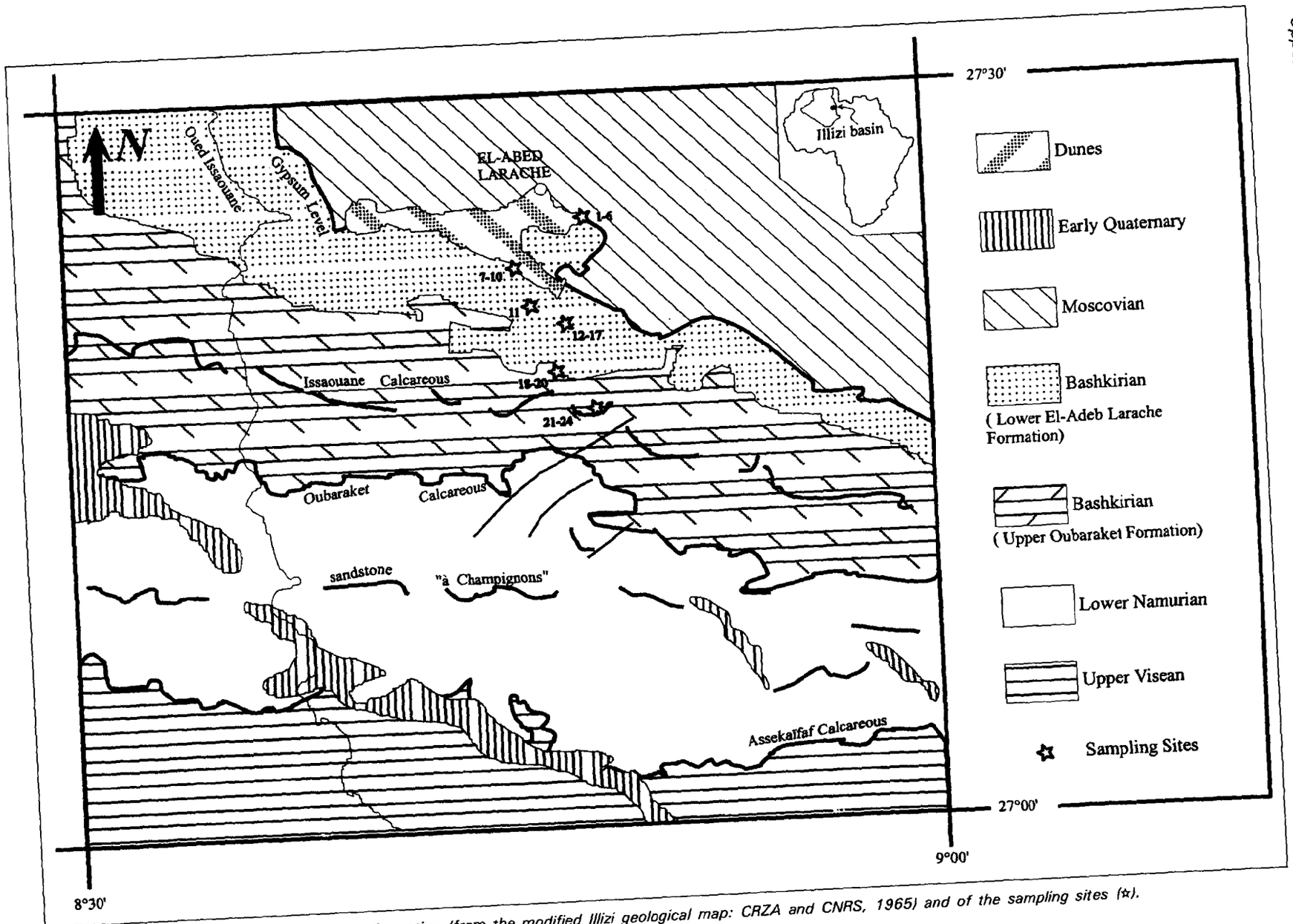


Figure 1. *Location of the studied Bashkirian Formation (from the modified Illizi geological map: CRZA and CNRS, 1965) and of the sampling sites (☆).*

ROCK MAGNETISM

The NRM intensities are variable and range from 6×10^{-4} to 2373×10^{-4} A m^{-1} (Fig. 2). For the demagnetised samples, the mean intensity is 192×10^{-4} A m^{-1}. Samples belonging to the red facies of the upper part of the formation exhibit the highest intensities. Blocking temperatures vary from site to site, ranging from 400° to 660°C (Fig 3).

Hysteresis loop measurements using a translation inductometer within an electromagnet were generated for specimens from many sites. Despite the dominance of a paramagnetic effect, these loops (Fig. 4) show that the saturation of the magnetisation is incomplete at 1.6 T and point

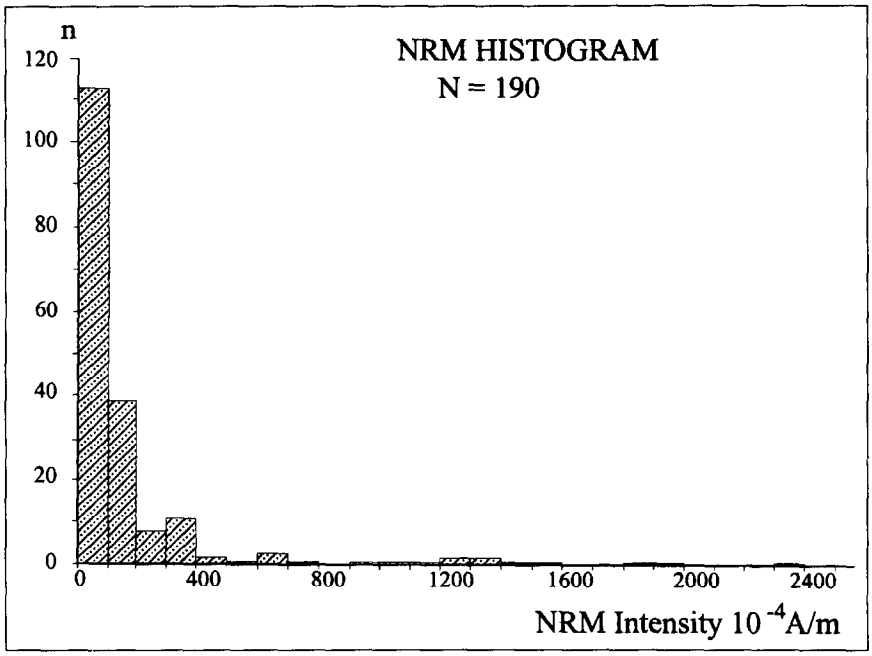


Figure 2. *NRM intensity versus number of samples (n) histogram. N: Total number of samples.*

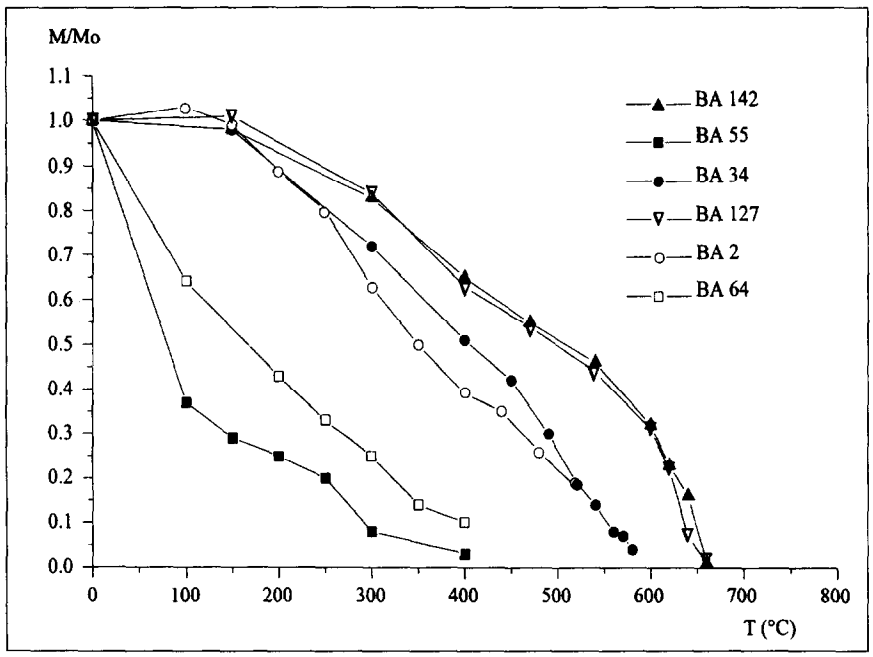


Figure 3. *Typical thermal demagnetisation curves of Bashkirian specimens BA2 (site 1), BA34 (site 2), BA55 (site 4), BA64 (site 5), BA127 (site 11) and BA142 (site 11) (see Table 1).*

out high remanent coercive forces (0.3–0.9 T) and relative low specific saturation magnetisations (0.1×10^{-3} to 2×10^{-3} A m^2 kg^{-1}). These data suggest hematite as the predominant carrier mineral.

The thermomagnetic curves indicate the existence of a single phase, with the Curie temperature higher than 600°C (Fig. 5). Hematite is the main magnetic carrier in these samples. In addition, the reversibility of the heating and cooling curves shows that this

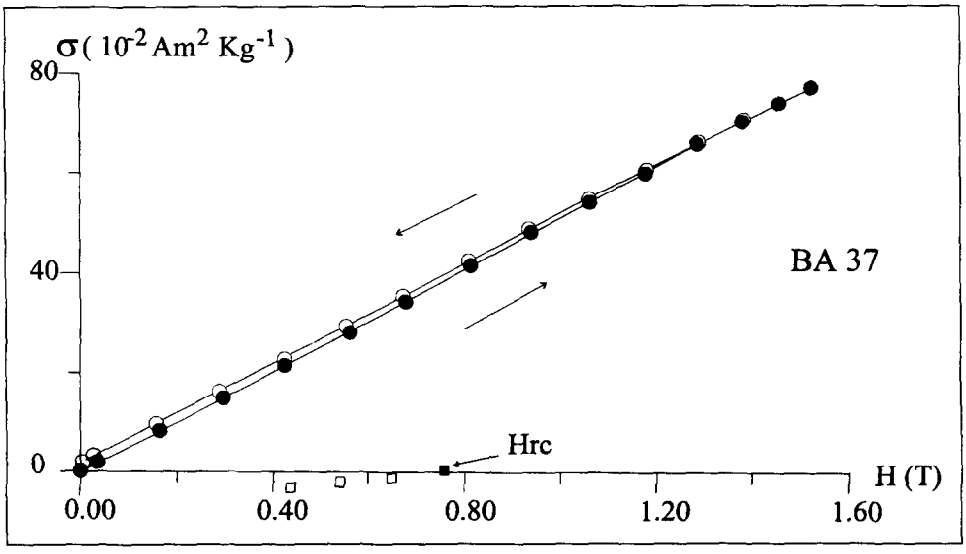


Figure 4. *Hysteresis curves: induced (●) and demagnetisation curve (○) of an Upper Bashkirian sample (Ba 37). Hrc: remanent coercive force; □: Hrc curve; ■: Hrc values; H: inducing field; T: tesla.*

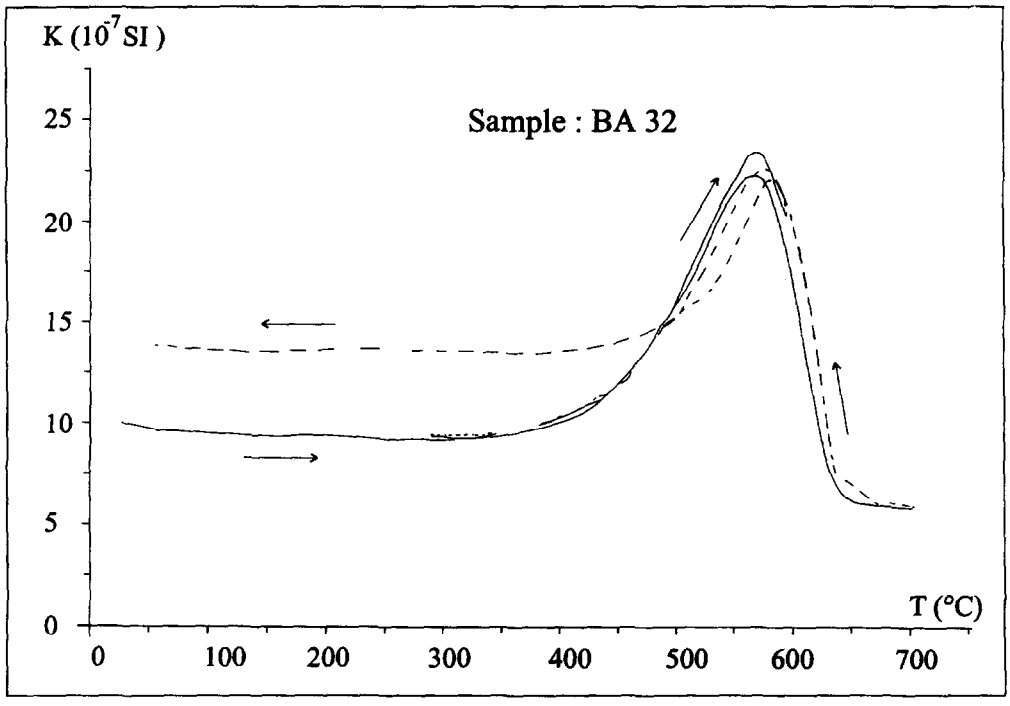


Figure 5. *Variation of the magnetic susceptibility during the cycle of progressive air heating (continuous line) and cooling (dashed line) of sample BA 32, showing, in addition to a 'Hopkinson' peak of hematite, the reversibility of the heating and cooling curves at different stages.*

hematite does not result from a mineralogical alteration during heating.

PALÆOMAGNETIC RESULTS

The NRM vector directions are distributed along a girdle (Fig. 6). This girdle is elongated between two clusters of directions, one along the Earth's present magnetic field direction and the other along a southeast down direction. This arrangement indicates the existence of two components in the NRM, at least for some of the samples.

During NRM analysis, a weak viscous component (A Component) was eliminated at the lowest temperature. Many specimens, corresponding to the clusters at the extremities of the girdle, show a stable magnetic direction (remaining vectors and/or vectorial differences) during the demagnetisation at higher temperatures. The ChRM, isolated after the analysis of such cases, presents three directions according to the sites (B component close to the present-day field, and C and D components in the southeast quadrant). In the other specimens, which have positions between these two extremes, the magnetisation direction evolves along a great circle path on the projection sphere.

ChRM of the secondary components (components B and C)

The B component (Table 1), isolated in one site (site 3), has a direction [declination (D) = 5.1°, inclination (I) = 26.3°, kappa (κ) = 38, alpha 95 (α_{95}) = 8.5°, number of samples (N) = 7: precision parameters k and α_{95} from Fisher, 1953] only slightly different than the Earth's present magnetic field at the site (D = -0.5°, I = 38°). It has been determined over a full range of temperatures, and is, therefore, not a viscous component, but rather a total overprint, which is likely to have had a chemical origin. Its corresponding virtual geomagnetic pole (VGP) (75.3°N, 171.5°E) is close to the palæomagnetic poles for other formations of the Illizi Basin and is associated with a Cenozoic magnetic overprint (Henry *et al.*, 1992; Derder *et al.*, 1994). This direction is interpreted, then, as a total remagnetisation that occurred during Cenozoic times.

The C component was isolated only in site 11 (D = 140.9°, I = 9.8°, κ = 148, α_{95} = 3.3°, N = 12; Table 1). This second orientation yielded a VGP (40.3°S, 64.3°E) that is very close to previous palæomagnetic poles, corresponding to Permian remagnetisations and determined in the overlying Moscovian Beds of the 'El-Adeb Larache Formation' (42°S, 65.1°E: Henry *et al.*, 1992) and in other areas of the Saharan Craton (35.5°S, 60°E at Hassi Bachir: Daly and Irving, 1983; 49.5°S, 42.2°E at Beni-Abbes:

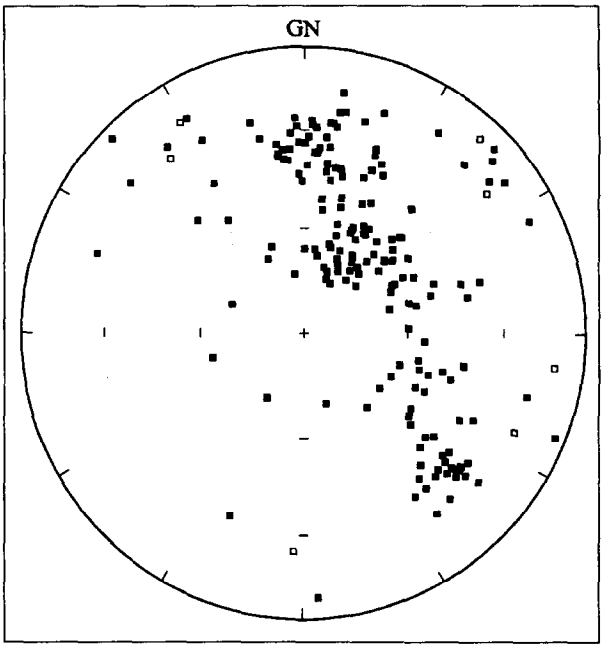


Figure 6. *Equal area plot of the NRM directions for all samples. □: negative inclinations; ■: positive inclinations.*

Aifa, 1987). Therefore, this component is considered to be a result of a remagnetisation, which occurred during Permian times.

ChRM of the D component

The D component (Fig. 7), observed in seven sites, has, after dip correction, the following direction (D = 137.3°, I = 34.1°, κ = 107, α_{95} = 5.1°, N = 7; Table 1). Since the magnetisation directions did not show any directional changes during demagnetisation and the demagnetisation curves did not reveal any suspect slope changes (Fig. 3), the data are not affected by parasitic effects due to heating. This has been confirmed by the good coherence of magnetisation directions within each site and between different sites (Fig. 8; Table 1).

Directions from great circle analysis

During the demagnetisation process, the magnetic direction of some specimens moved along great circles, starting near the present-day field (PDF) direction, evolving towards the southeast (Fig. 9) and ending before the intersection of the different great circles (McFadden and McElhinny, 1988). The analysis of remagnetisation circles was first studied using the method of Halls (1976, 1978). The obtained intersection directions appear relatively similar for the different sites (Table 2) in spite of the weak number of available specimens (3 or 4) for each site. A good correlation exists between these intersections and the ChRM directions determined

Table 1. Site number and mean directions obtained after dip correction in each site from Bashkirian ChRM directions

Site	N	D	I	k	α_{95}	Lat. (S)	Long. (E)	K	A_{95}
Recent Overprint									
3	7	5.1	26.3	38	8.5	-75.6	171.5	44	8.0
Permian Overprint									
11	12	140.9	9.8	148	3.3	40.3	64.3	183	3.0
Bashkirian D[tions]									
1	26	136.0	26.5	116	2.5	30.6	60.0	153	2.2
2	13	134.9	28.3	178	2.9	29.1	60.0	210	2.7
5	3	133.7	39.2	106	7.8	23.3	55.3	95	8.3
6	4	137.4	30.7	69	8.4	29.5	57.0	76	8.1
8	2	141.9	30.6			32.8	53.2		
12	5	140.3	48.7	476	2.9	21.7	45.0	309	3.6
13	2	145.8	40.1			30.3	44.1		
Mean 55 specimens	55	136.6	30.6	64	2.3	29.2	57.8		
Mean 7 sites		137.3	34.1	107	5.1	28.1	55.5		
Mean 7 sites						28.3	53.5	130	4.6

Palæomagnetic pole: latitude (°S), longitude (°E), and corresponding Fisher's (1953) parameters: k, α_{95}, K, A_{95} (α_{95}, A_{95}: in degrees); D, I: in degrees; N: number of samples in each site.

in other samples from the same sites (Table 1), except for site 11, where the ChRM corresponds to a Permian overprint. In this site, the great circles intersection has a similar orientation as in the other sites. This shows that the samples with Permian magnetic overprint were wholly remagnetised, but that the other samples were not significantly affected by this Permian remagnetisation. A similarity also exits between the intersection direction using all remagnetisation circles (Fig. 10; Table 2) and the ChRM mean direction of component D (using 55 specimens; Table 1).

To obtain a more reliable result that includes a zone of confidence, the McFadden and McElhinny (1988) approach is usable to integrate great circles and ChRM data (without ChRM for site 11). The results obtained by this method are in good agreement with those using the method of Halls (1976, 1978) (Table 2). This confirms that one of the components revealed by the scattering of the directions along great circles is the ChRM defined earlier.

Age of the D component

From the above, it is clear that the ChRM data are consistent with those obtained by remagnetisation circle analysis (Tables 1 and 2). After dip correction, the mean ChRM direction obtained by giving unit weight to each specimen ($D = 136.6°$, $I = 30.6°$, $\kappa = 64$, $\alpha_{95} = 2.3°$, $N = 55$) is very close to that obtained by giving unit weight to each of the seven sites ($D = 137.3°$, $I = 34.1°$, $\kappa = 107$, $\alpha_{95} = 5.1°$). When integrating the great circle data (McFadden and McElhinny, 1988), the mean direction for eight sites after dip correction was $D = 137.5°$, $I = 33.7°$, $\kappa = 118$, $\alpha_{95} = 4.5°$; sites 1, 8, 12 and 13 used the ChRM and great circles data; sites 2, 5 and 6 used only ChRM data; and site 11 used only the great circle data.

Slight variations of directions were found between stratigraphic levels. In particular, the steepest inclinations were observed in the oldest level (sites 12 and 13; see Table 1), as expected for primary components during this period in this area (from APWP analysis). This suggests that the observed direction does not result from a regional magnetic overprint. It can be seen that:

i) the observed direction is obtained from fine-grained specimens with a low degree of weathering;

ii) the rocks belong to a stable area (no evidence of deformation effects or volcanic activity);

iii) this direction is different from that of the Permian overprint described from site 11 and other areas of the Sahara Craton (see paragraph A);

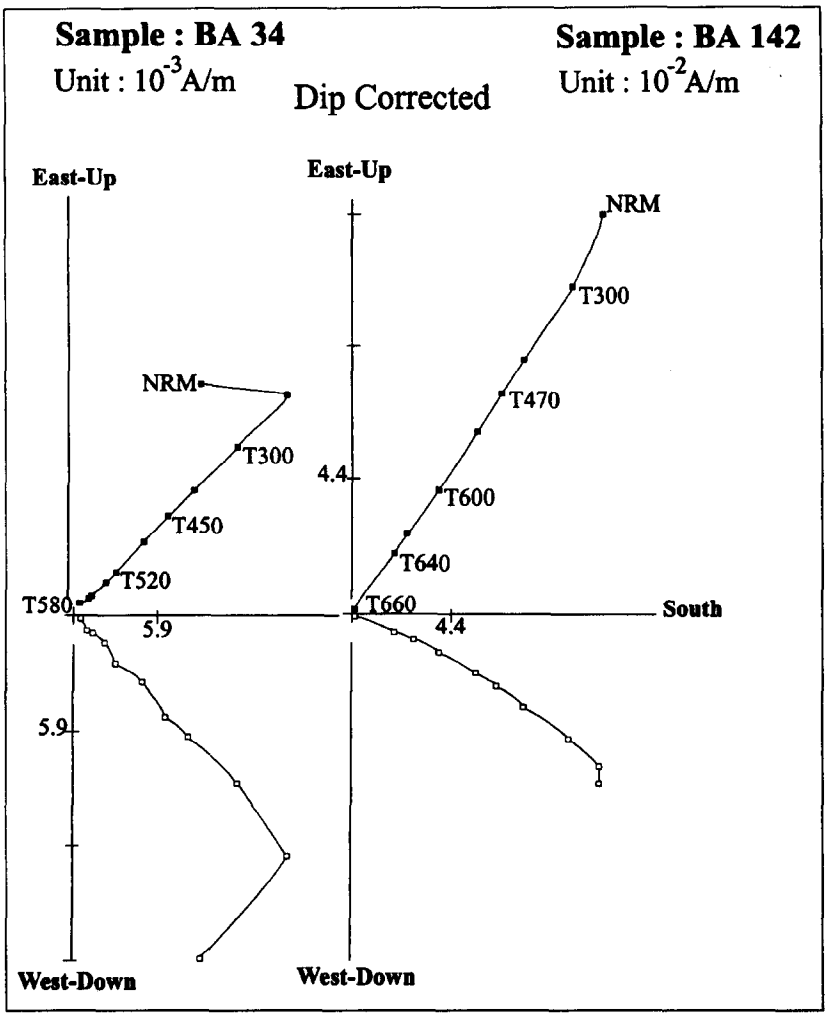


Figure 7. Evolution of the magnetisation vector during progressive thermal demagnetisation. Projection on horizontal (□) and on vertical (■) planes (orthogonal vector plot) of samples BA 34 and BA142 showing a stable magnetic direction. NRM: natural remanent magnetisation; T: temperature of demagnetisation step, in °C.

Table 2. Site number and mean direction obtained after dip correction from each site using remagnetisation circles analysis (Halls, 1978; McFadden and McElhinny, 1988) methods

			Halls method		*McFadden and McElhinny method*		
Site	**N**	**N'**	**D**	**I**	**D**	**I**	α_{95}
1	26	4	133.9	34.1	136.1	26.6	2.3
8	2	3	130.6	38.4	138.0	30.1	7.0
11	-	4	136.8	30.8	136.8	30.8	-
12	5	3	140.9	41.2	138.9	46.1	2.7
13	2	3	145.1	29.3	143.1	35.5	15.4
Total	34	17	136.7	33.7	137.5	31.0	2.5
Mean 5 sites			137.5	34.9	138.5	34.2	6.7

D, I, α_{95}: in degrees; N: number of ChRM; N': number of circles.

iv) this direction did not result from a regional overprint; and

v) this direction is close to that determined from the overlying and well-dated Moscovian Beds of the 'El-Adeb Larache Formation' (Henry *et al.*, 1992).

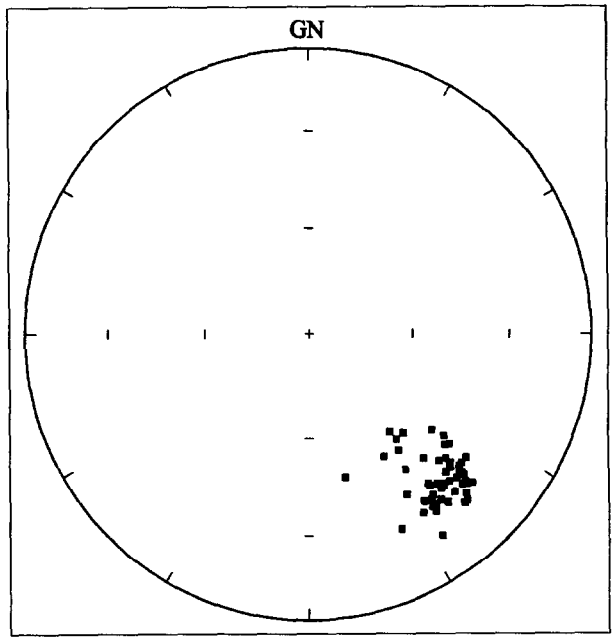


Figure 8. *Equal area plot of the ChRM of Bashkirian Beds. ■: positive inclinations.*

It can, therefore, be assumed that the component D represents a magnetisation of Bashkirian age.

The palæomagnetic pole associated with this component (28.2°S, 55.5°E, K = 207, A_{95} = 3.4°; where K and A_{95} are precision parameters of Fisher, 1953) is very close to those obtained from the neighbouring periods, the Moscovian Pole (28.7°S, 55.9°E: Henry *et al.*, 1992) at El-Adeb Larache, the Upper Namurian-Lower Moscovian Pole (26.8°S, 56.6°E: Daly and Irving, 1983) at Hassi Bachir, the Lower Moscovian Pole (22.9°S, 51.8°E: Daly and Irving, 1983) at Aïn Ech Chebbi and the Moscovian Pole (24.0°S, 62.1°E; Derder *et al.*, *in press*) at Edjeleh.

Since it was obtained from several levels of the Upper and Lower Bashkirian, it, therefore, constitutes a new palæomagnetic pole, which specifies the African APWP.

GEODYNAMICAL IMPLICATIONS

The large difference in pole position, existing between the Upper Devonian and the Upper Carboniferous, was previously interpreted as due to a large clockwise rotation of Gondwana before Moscovian times (Henry *et al.*, 1992). The agreement of the results of this study with the results of the previous African Upper Carboniferous poles shows that this displacement ended before or during the Bashkirian.

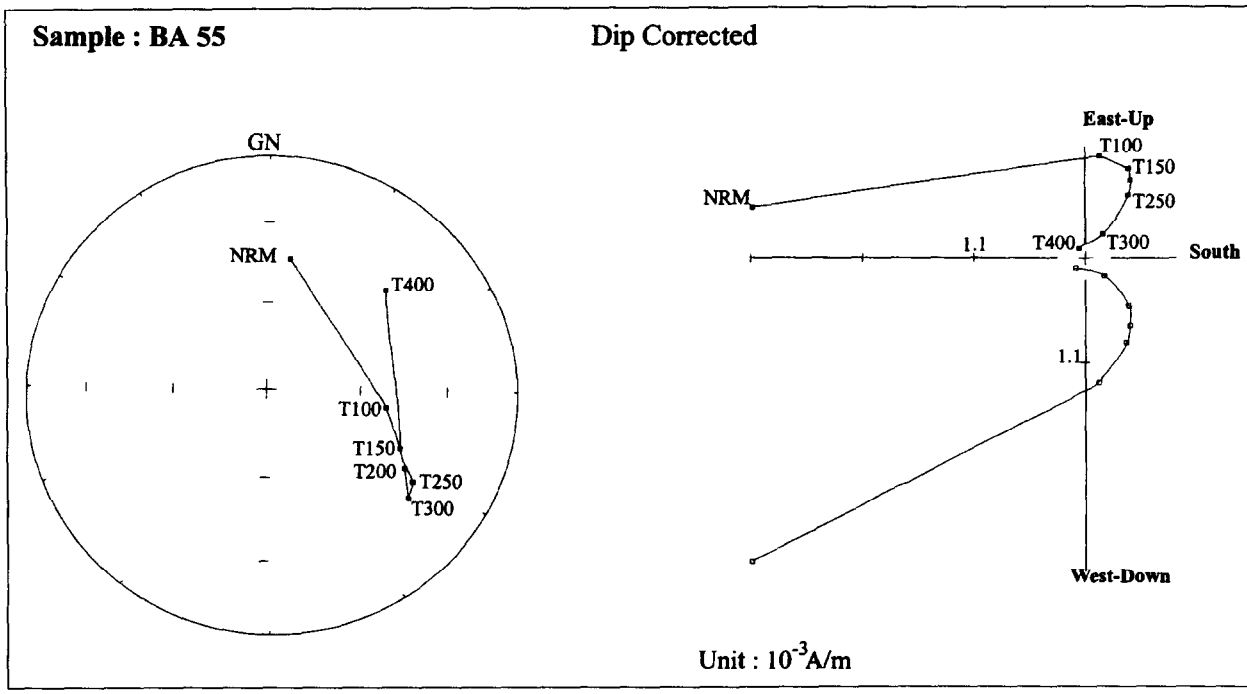


Figure 9. *Evolution of the magnetisation vector during progressive thermal demagnetisation. Projection on horizontal (□) and on vertical (■) planes (remaining vectors) of sample BA 55 showing a superposition of unblocking temperature spectra. T: temperature of demagnetisation step, in °C.*

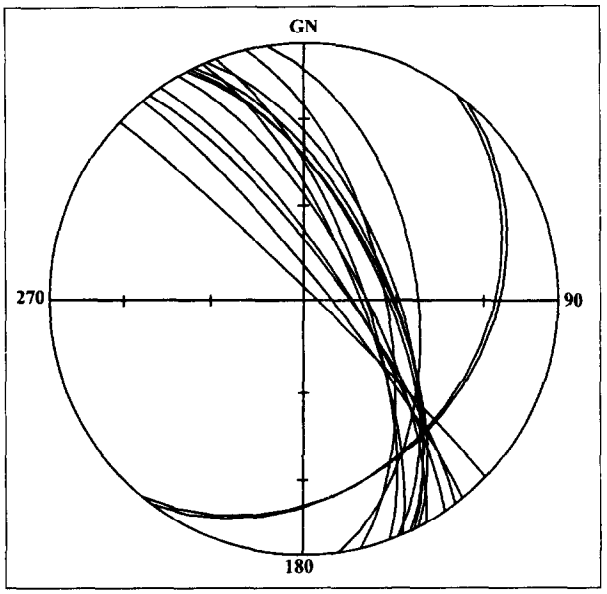


Figure 10. *Best estimation intersection of remagnetisation circles obtained from 17 specimens (Table 2) distributed over five sites (equal area plot, lower hemisphere).*

It also confirms the good quality of the previous Upper Carboniferous poles for Africa. Using a classical reconstruction (Fig. 11: Lottes and Rowley, 1990) derived from that of du Toit (1937), this pole is consistent with the poles from South America and Madagascar (Western Gondwana) but differs from Australia poles (Eastern Gondwana). In order to compare Western and Eastern Gondwana poles, the selection of Carboniferous poles of Chen *et al.*, (1994) (Figs 11 and 12) was used. Some poles (Chen *et al.*, 1994, mnemonic FBC, BO, SO, MES and BC in Australia, TA in India, and BO and GC in Africa) that have very different orientations from the other poles (perhaps because of magnetic overprinting or poor age constraints) were rejected in the mean pole in this study. The angular difference between Western and Eastern Gondwana mean poles appears significant using the Lottes and Rowley (1990) reconstruction, 20° (Fig. 11). However, an alternative reconstruction proposed by Ricou *et al.* (1990) gives an angular difference of

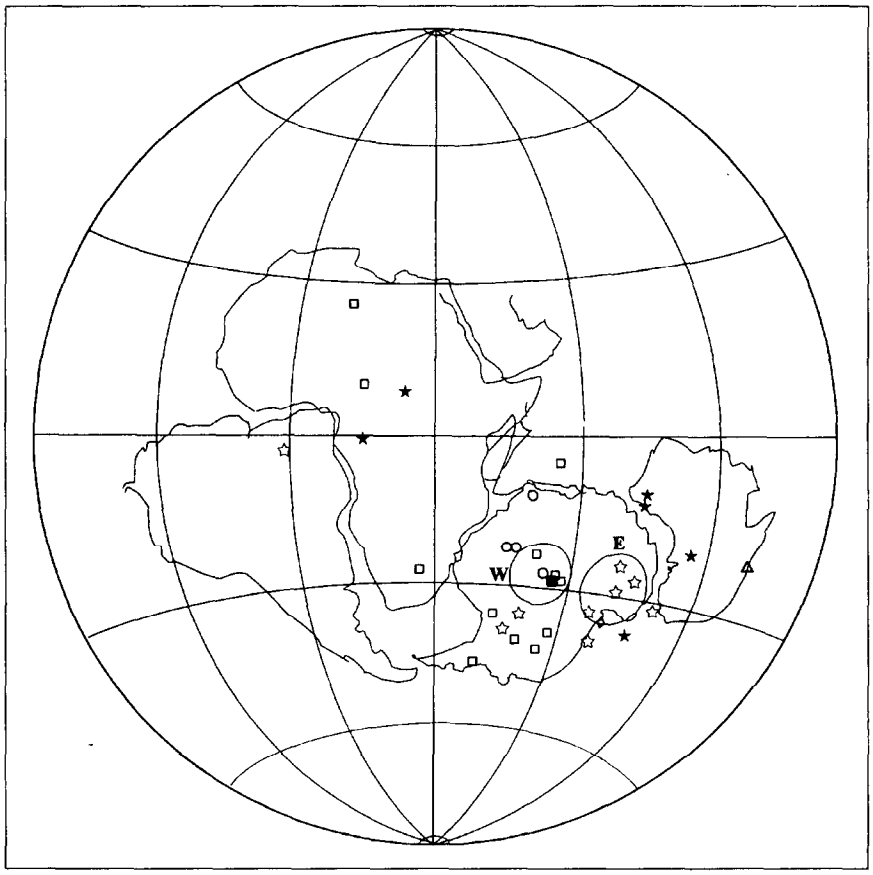


Figure 11. *Carboniferous palæomagnetic poles (Chen* et al., *1994) from Gondwana in the Lottes and Rowley (1990) reconstruction (Africa fixed) with a confidence zone for mean poles for Western (W) and Eastern (E) Gondwana. ★: Cratonic Australia; ☆: Southeastern Australia; △: India; ◇:Madagascar; ○: South America; □: Africa; ■: Africa, the new Bashkirian Pole from this study.*

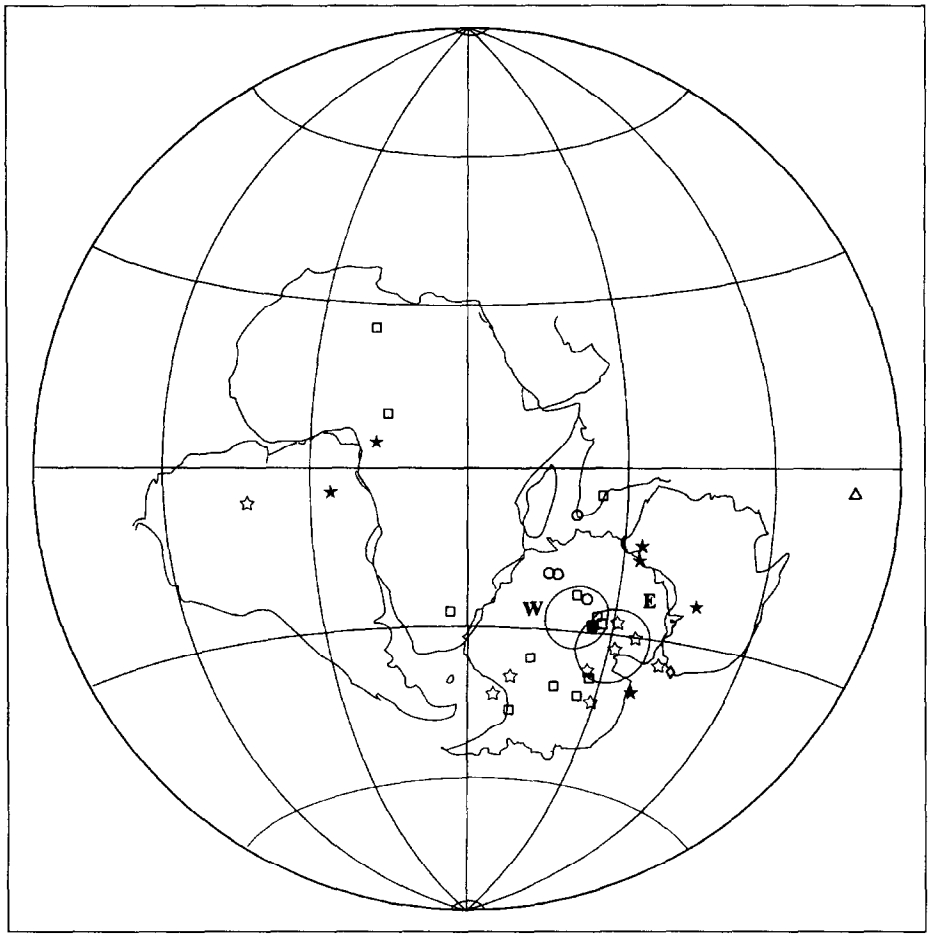


Figure 12. *Carboniferous palæomagnetic poles from Gondwana in Ricou* et al. *(1990) reconstruction (see Fig. 11 for symbol definitions).*

only 6° with intersecting confidence zones for the mean poles (Fig. 12). Therefore, it clearly represents a better solution than that of the classical reconstruction for this time interval.

CONCLUSION

This palæomagnetic study yields two palæomagnetic poles. The first (40.3°S, 64.2°E) is attributed to a Permian remagnetisation; the second pole (28.2°S, 55.5°E, $K = 207$, $A_{95} = 3.4°$) is considered to be primary and of Bashkirian age. This second pole is in good agreement with the previously published Saharan poles for neighbouring periods. The comparison of this new datum with the previous Carboniferous Gondwana poles shows that a better reconstruction is obtained for Gondwana if the parameters of Ricou *et al.* (1990) are used rather those of the classical model of du Toit (1937). However, an angular difference between the Western and Eastern Gondwana mean pole still exists, and more studies are needed to refine the Gondwana reconstruction.

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